Sampling Distributions: Mean & Standard Deviation

Statistics Worksheet · Grade 11–12 / Introductory College

Name: _____ Date: _____

Learning Objectives

- Apply the Central Limit Theorem to identify the mean of a sampling distribution
- Calculate the standard deviation (standard error) of a sampling distribution using the formula σ/√n
- Solve probability questions involving sampling distributions using z-scores and the normal curve

Problems

1. A population has a mean of 50. According to the Central Limit Theorem, what is the mean of the sampling distribution of sample means drawn from this population?

$$\mu_{\bar{x}} = ?$$

2. A population has a standard deviation of 12. A random sample of size 36 is drawn. Find the standard deviation of the sampling distribution (standard error).

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}}$$

3. The heights of young women are normally distributed with a population mean of 64.5 inches and a population standard deviation of 2.5 inches. If 10 women are randomly selected, what is the mean of the sampling distribution?

$$\mu_{\bar{x}} = ?$$

4. Using the same women's height distribution (mean = 64.5 inches, standard deviation = 2.5 inches), find the standard deviation of the sampling distribution when 10 women are randomly selected. Round to two decimal places.

$$\sigma_{\bar{x}} = \frac{2.5}{\sqrt{10}}$$

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5. Using the women's height distribution (mean = 64.5 inches, standard deviation = 2.5 inches), compute the standard deviation of the sampling distribution when the sample size increases to 100. How does this compare to the result for n = 10?

$$\sigma_{\bar{x}} = \frac{2.5}{\sqrt{100}}$$

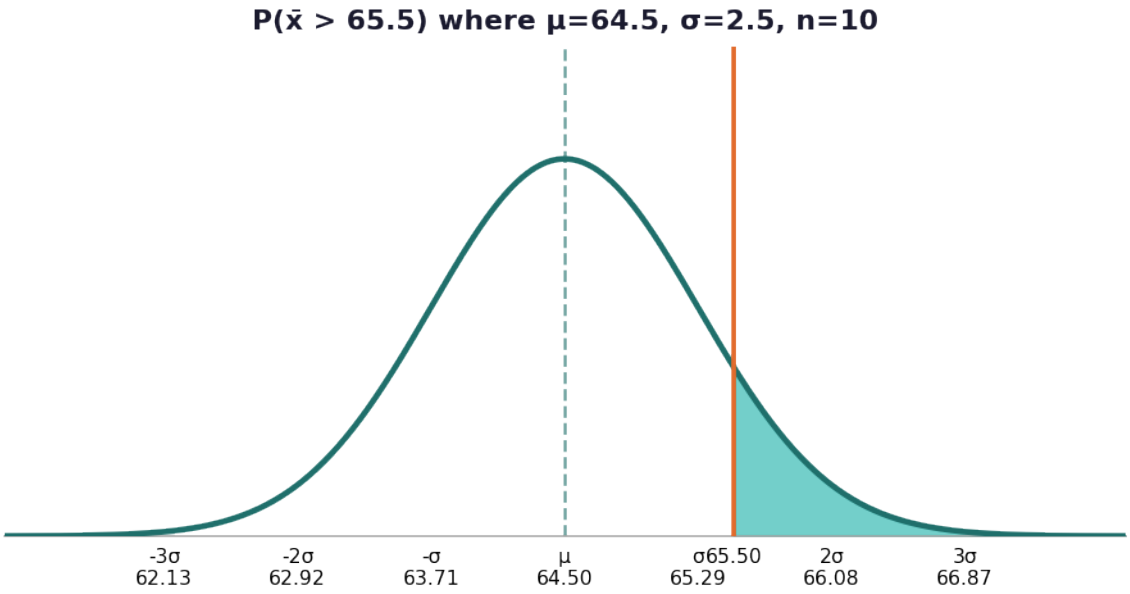
6. Fill in the missing values in the table below for the women's height distribution (μ = 64.5, σ = 2.5). Round standard errors to two decimal places.

Sample Size (n)	Sample Mean (μ_x̄)	Standard Error (σ_x̄)
10	64.5	
25	64.5	
100	64.5	
400	64.5	

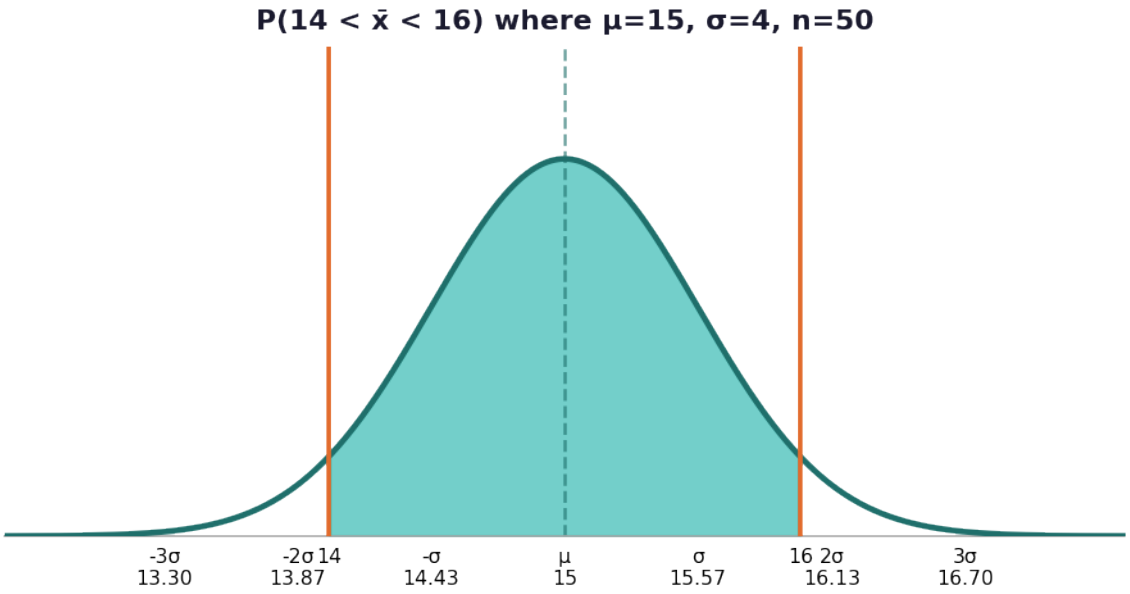
7. A manufacturer fills juice bottles with a mean of 16 oz and a standard deviation of 0.8 oz. A quality inspector randomly selects 64 bottles. Find the mean and standard deviation of the sampling distribution of the sample mean.

$$\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

8. The heights of young women are normally distributed with a mean of 64.5 inches and a standard deviation of 2.5 inches. A sample of 10 women is randomly selected. Find the probability that the sample mean height is greater than 65.5 inches. Use the standard normal distribution.

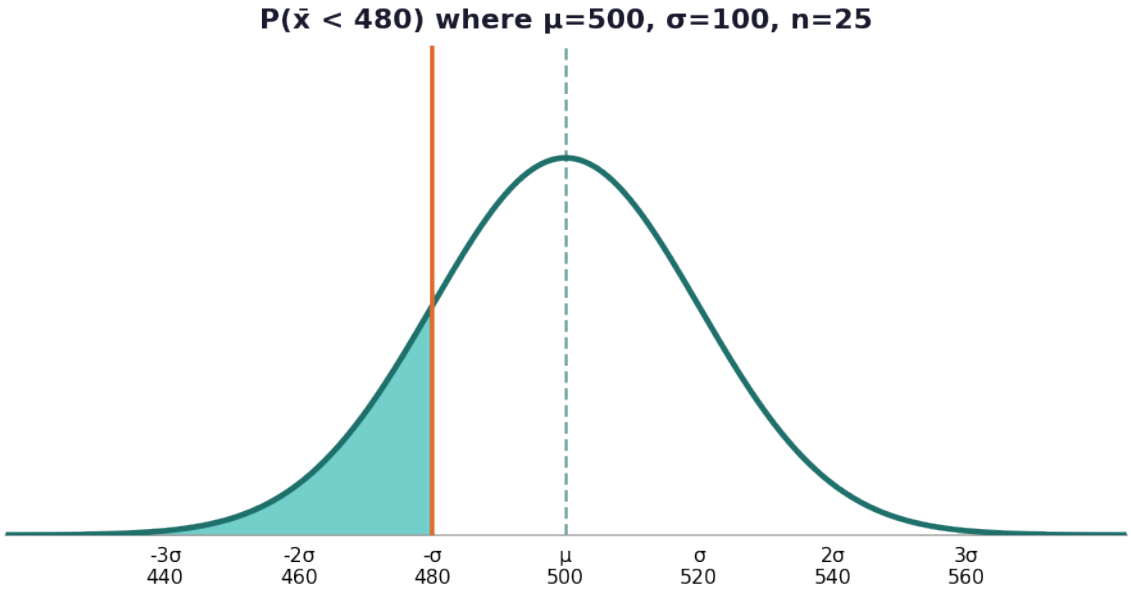


9. A university reports that the average student studies 15 hours per week with a standard deviation of 4 hours. A random sample of 50 students is taken. What is the probability that the sample mean study time is between 14 and 16 hours per week?



10. Scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. A researcher draws a random sample of 25 students. (a) Find the mean and standard error of the

researcher uses a sample of 100 students instead. Without computing, explain how this changes the standard error and the probability in part (b).



Sampling Distributions: Mean & Standard Deviation — Answer Key

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Answer Key

1. Answer: $\mu_{\bar{x}} = 50$

- The Central Limit Theorem states that the mean of the sampling distribution equals the population mean.
- Therefore $\mu_{\bar{x}} = \mu = 50$.

2. Answer: $\sigma_{\bar{x}} = 2$

- Use the formula $\sigma_{\bar{x}} = \sigma / \sqrt{n}$.
- $\sigma_{\bar{x}} = 12 / \sqrt{36} = 12 / 6 = 2$.

3. Answer: $\mu_{\bar{x}} = 64.5$ inches

- By the Central Limit Theorem, the sampling distribution mean equals the population mean.
- Therefore $\mu_{\bar{x}} = 64.5$ inches.

4. Answer: $\sigma_{\bar{x}} \approx 0.79$ inches

- Use $\sigma_{\bar{x}} = \sigma / \sqrt{n} = 2.5 / \sqrt{10}$.
- $\sqrt{10} \approx 3.162$, so $\sigma_{\bar{x}} = 2.5 / 3.162 \approx 0.79$ inches.

5. Answer: $\sigma_{\bar{x}} = 0.25$ inches; smaller than the n=10 result of 0.79 inches

- $\sigma_{\bar{x}} = 2.5 / \sqrt{100} = 2.5 / 10 = 0.25$ inches.
- For n = 10, $\sigma_{\bar{x}} \approx 0.79$ inches. As sample size increases, the standard error decreases.

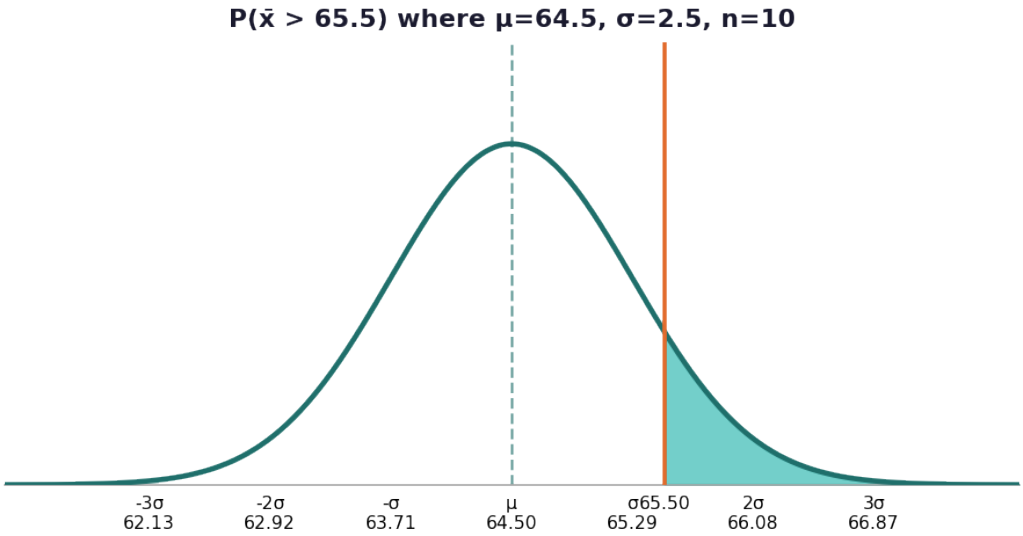
6. Answer: See completed table

Sample Size (n)	Sample Mean ($\mu_{\bar{x}}$)	Standard Error ($\sigma_{\bar{x}}$)
10	64.5	0.79
25	64.5	0.50
100	64.5	0.25
400	64.5	0.13

- For each row, apply $\sigma_{\bar{x}} = 2.5 / \sqrt{n}$.
- n=10: $2.5/\sqrt{10} \approx 0.79$ | n=25: $2.5/\sqrt{25} = 0.50$ | n=100: $2.5/\sqrt{100} = 0.25$ | n=400: $2.5/\sqrt{400} = 0.13$.

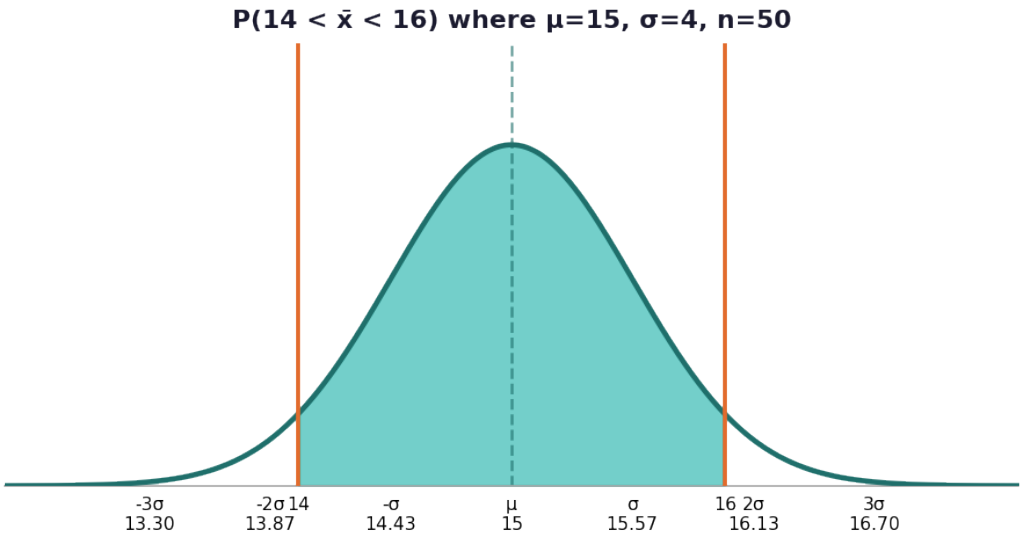
7. Answer: $\mu_{\bar{x}} = 16$ oz, $\sigma_{\bar{x}} = 0.1$ oz

- Mean of sampling distribution: $\mu_{\bar{x}} = \mu = 16$ oz.
- Standard error: $\sigma_{\bar{x}} = 0.8 / \sqrt{64} = 0.8 / 8 = 0.1$ oz.



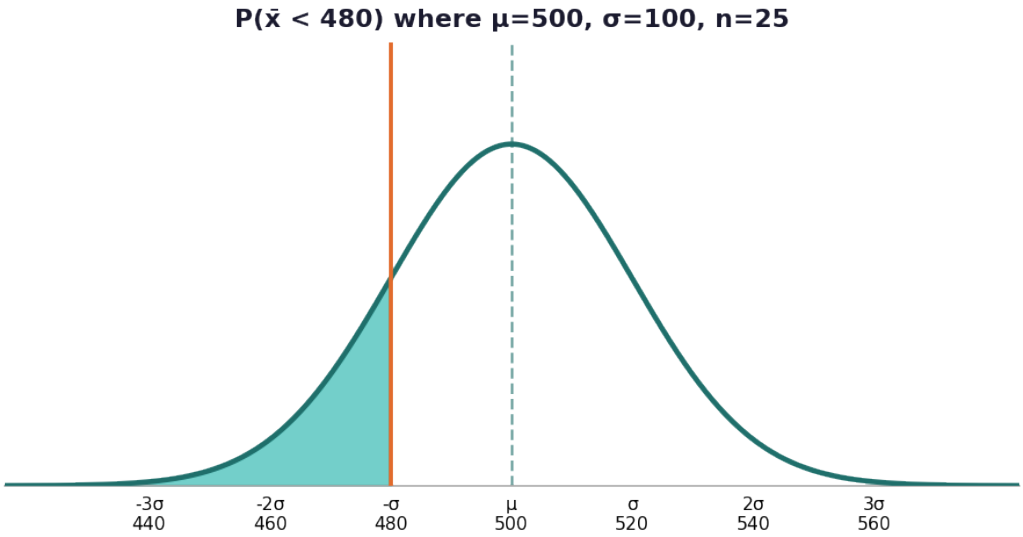
- Step 1: Find $\sigma_{\bar{x}} = 2.5 / \sqrt{10} \approx 0.79$ inches.
- Step 2: Compute $z = (65.5 – 64.5) / 0.79 = 1.0 / 0.79 \approx 1.27$.
- Step 3: $P(z > 1.27) = 1 – P(z < 1.27) = 1 – 0.8973 \approx 0.1027$.
- The probability that the sample mean exceeds 65.5 inches is approximately 10.27%.

9. Answer: P ≈ 0.9216 (92.16%)



- Step 1: $\sigma_{\bar{x}} = 4 / \sqrt{50} \approx 0.566$ hours.
- Step 2: $z_1 = (14 – 15) / 0.566 \approx –1.77$ and $z_2 = (16 – 15) / 0.566 \approx 1.77$.
- Step 3: $P(–1.77 < z < 1.77) = P(z < 1.77) – P(z < –1.77) = 0.9616 – 0.0384 \approx 0.9232$.
- The probability that the sample mean is between 14 and 16 hours is approximately 92.32%.

10. Answer: (a) $\mu_{\bar{x}} = 500$, $\sigma_{\bar{x}} = 20$; (b) P ≈ 0.1587 (15.87%); (c) $\sigma_{\bar{x}}$ drops to 10, making it less likely to get $\bar{x} < 480$



- (a) $\mu_{\bar{x}} = 500$ (equals population mean); $\sigma_{\bar{x}} = 100 / \sqrt{25} = 100 / 5 = 20$.
- (b) $z = (480 – 500) / 20 = –20 / 20 = –1.00$.
- $P(z < –1.00) = 0.1587$, so the probability is approximately 15.87%.
- (c) With $n = 100$, $\sigma_{\bar{x}} = 100 / \sqrt{100} = 10$. The sampling distribution is narrower.
- The z-score becomes $(480 – 500) / 10 = –2.00$, giving $P \approx 0.0228$ — much smaller probability.